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The Capex/Opex Inflection: When Bare Metal Beats Cloud Routing

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Abstract

The cost-economics debate around agentic operations has converged on a slogan: cloud is cheaper at small scale; on-premises is cheaper at large scale. Like most slogans about complex economic decisions, it is approximately wrong in operationally consequential ways. The actual inflection between cloud opex and bare-metal capex is workload-class dependent, sensitive to a half-dozen parameters that the slogan obscures, and crossable in both directions over a system's operating lifetime as workload characteristics, accelerator generation, and capacity-sourcing posture evolve.

This paper specifies the crossover mathematics, identifies five workload classes with distinct inflection profiles, and treats the cost decision as a governance-layer concern within the HGC³AE² framework rather than as a finance-only or infrastructure-only decision. The argument is that mission-ready agentic operation has a workload mix dominated by classes whose inflection sits much earlier than the cloud-default slogan suggests, and that organizations applying the slogan uniformly are systematically over-paying for cloud opex on workloads that capex amortization would carry more cheaply.

This is Paper 3 of the *Managing Agentics Ops* series, closing the foundation triad with Paper 1 (Capacity Sourcing) and Paper 2 (Heterogeneous Compute Routing). The remainder of the series develops operational depth on the substrate these three papers establish.

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ABSTRACT

The cost-economics debate around agentic operations has converged on a slogan: *cloud is cheaper at small scale; on-premises is cheaper at large scale*. Like most slogans about complex economic decisions, it is approximately wrong in operationally consequential ways. The actual inflection between cloud opex and bare-metal capex is **workload-class dependent**, sensitive to a half-dozen parameters that the slogan obscures, and crossable in both directions over a system's operating lifetime as workload characteristics, accelerator generation, and capacity-sourcing posture evolve.¹²

This paper specifies the crossover mathematics, identifies five workload classes with distinct inflection profiles, and treats the cost decision as a governance-layer concern within the HGC³AE² framework rather than as a finance-only or infrastructure-only decision. The argument is that mission-ready agentic operation has a workload mix dominated by classes whose inflection sits much earlier than the cloud-default slogan suggests, and that organizations applying the slogan uniformly are systematically over-paying for cloud opex on workloads that capex amortization would carry more cheaply.

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1. INTRODUCTION

The cost-economics decision in cloud computing has been treated, in the practitioner literature and increasingly in peer-reviewed work, as a total-cost-of-ownership (TCO) calculation comparing per-unit cloud pricing against amortized capex for an equivalent on-premises footprint.³ The standard treatment is correct as far as it goes, but it conceals three structural facts that determine where the inflection actually sits for any given organization.

The first concealed fact is that **the inflection is workload-class dependent**. Cloud opex prices in elasticity headroom that bursty workloads consume; bare-metal capex prices in committed utilization that steady-state workloads consume. For workload classes whose demand profile is genuinely bursty, opex per unit beats amortized capex per unit; the elasticity discount is real and operationally valuable. For workload classes whose demand profile is steady-state, opex per unit loses to amortized capex per unit; the elasticity discount is paid for

1 Frank Pallas et al., "The Economics of Cloud Computing: A Survey," *Future Generation Computer Systems* 115 (2021): 78–96, <https://doi.org/10.1016/j.future.2020.08.019>.

2 Eric Masanet et al., "Recalibrating Global Data Center Energy-Use Estimates," *Science* 367, no. 6481 (2020): 984–986, <https://doi.org/10.1126/science.aba3758>.

3 Joe Weinman, *Cloudonomics: The Business Value of Cloud Computing* (Hoboken, NJ: Wiley, 2012).



but not used. Agentic operating workloads tilt heavily toward steady-state — sustained inference, persistent memory operations, governance overhead, retrieval-augmented generation — which shifts the inflection materially earlier than the cloud-default slogan suggests.⁴

The second concealed fact is that **the inflection's parameters are not stable over time**. Accelerator generation cycles, regional power-price evolution, sovereignty-regime changes, and the operator's own scale all move the inflection. An organization that crossed the inflection two years ago in a particular direction may find, today, that the parameters have shifted and the inflection has crossed back. The decision is not once-and-done; it is a continuous governance concern.⁵

The third concealed fact is that **the inflection has structural exceptions**. For DDIL-tolerant forward-deployed workloads (MR-P4 cognitive nodes), the inflection is not economic; it is structural. Sovereignty, latency, and connectivity make bare-metal on-premises capacity mandatory regardless of which side of the economic inflection the workload would otherwise sit.⁶ The paper treats this exception explicitly in §5.

The paper proceeds in the canonical Mission-Ready shape. Section 2 names the failure modes of single-axis cost reasoning. Section 3 specifies the workload-class taxonomy and the crossover math. Section 4 treats the inflection as a continuous concern. Section 5 covers the DDIL structural exception. Section 6 returns to HGC³AE² and develops operator implications.

2. PROBLEM SET: FIVE FAILURE MODES OF SINGLE-AXIS COST REASONING

Single-axis cost reasoning — *cheaper per hour wins* or *amortized capex per hour wins* — fails in five structurally distinct ways.

2.1 SLOGAN-DRIVEN DECISIONS THAT SURVIVE THEIR OWN PARAMETER DRIFT

Organizations that have adopted the *cloud is cheaper at small scale, on-prem at large scale* heuristic make the decision once and treat it as durable. The parameters that determined the original decision — accelerator generation, regional power prices, sovereignty regime, workload mix — drift, sometimes substantially, over the typical operating window. The decision does not drift with them. The organization continues to pay the cloud opex premium long after the workload mix has shifted to steady-state, or continues to operate amortized on-prem capacity long after the workload has shifted to genuinely bursty. The failure is structural inflexibility against parameter drift.⁷

4 Christina Delimitrou and Christos Kozyrakis, “Quasar: Resource-Efficient and QoS-Aware Cluster Management,” *Proceedings of ASPLOS '14*, 2014, <https://doi.org/10.1145/2541940.2541941>.

5 Lotfi Mhamdi, Mohamed Mohsen Ben Khelifa, and Pascale Vicat-Blanc Primet, “Power Consumption Models in Cloud Computing: A Survey,” *Journal of Network and Computer Applications* 159 (2020): 102591, <https://doi.org/10.1016/j.jnca.2020.102591>.

6 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

7 Nikolas Roman Herbst, Samuel Kounev, and Ralf Reussner, “Elasticity in Cloud Computing: What It Is, and What It Is Not,” *Proceedings of the 10th International Conference on Autonomic Computing (ICAC '13)*, 2013, <https://www.usenix.org/conference/icac13/technical->



2.2 AGGREGATE-TCO CALCULATIONS THAT HIDE CLASS-SPECIFIC INFLECTIONS

A common TCO calculation sums all workload across a uniform per-unit price and compares aggregate cost to aggregate amortized capex. The aggregate calculation hides the class-specific inflections that the workload mix actually contains. A workload mix that is 70% steady-state and 30% bursty produces an average per-unit cost that obscures the fact that the 70% should route to capex-amortized substrate and the 30% should route to opex-elastic substrate. The aggregate calculation tells the organization to choose one substrate or the other; the actual answer is to route each class to the substrate where its inflection favors it.⁸

2.3 COST OPTIMIZATION AGAINST PRICE SIGNALS WITHOUT COST STRUCTURE

A second related failure is optimizing against the price signal each cloud provider exposes without reasoning about the cost structure that produces the signal. A spot-price discount, sustained-use commitment, or reserved-instance offer reflects the provider's own upstream sourcing position (MAO-P1) and its margin-management decisions; it does not reflect a durable change in the operator's cost structure.⁹ Reorganizing the operating model around a short-term price signal produces architectural lock-in that the next price-signal change must reverse, often at greater cost than the original optimization saved.

2.4 CAPEX DECISIONS MADE WITHOUT OPERATIONAL-LIFECYCLE COST AWARENESS

Bare-metal capex decisions made on per-accelerator MSRP, without modeling the operational-lifecycle costs (power, cooling, ops staffing, refresh cycles, decommissioning, accelerator generation rollover), produce capex-vs-opex comparisons that overstate capex's advantage. The lifecycle cost is comparable in magnitude to the capex itself for hyperscale operations; ignoring it produces inflection estimates that are systematically off in the on-prem-favorable direction.¹⁰

2.5 GOVERNANCE-LAYER DETACHMENT FROM COST DECISIONS

Cost decisions of this scale routinely sit with infrastructure or platform teams whose accountability is operational rather than strategic. The governance-layer participants who would otherwise review sovereignty, allocation, and capacity envelopes are absent from the cost decision; the result is cost-optimization decisions that breach sovereignty envelopes, exit strategic-allocation commitments, or exhaust capacity-headroom commitments without governance-layer awareness. The decision is locally rational and globally incoherent.

sessions/presentation/herbst.

8 Vinod Kumar Vavilapalli et al., "Apache Hadoop YARN: Yet Another Resource Negotiator," *Proceedings of SoCC '13*, 2013, <https://doi.org/10.1145/2523616.2523633>.

9 Jonatha Anselmi and Bruno Gaujal, "Pricing the Cloud: Resource Allocations, Fairness, and Revenue," *Performance Evaluation* 119 (2018): 1–15, <https://doi.org/10.1016/j.peva.2017.12.001>.

10 Eric Masanet et al., "Recalibrating Global Data Center Energy-Use Estimates," *Science* 367, no. 6481 (2020): 984–986, <https://doi.org/10.1126/science.aba3758>.



2.6 THE UNIFIED PROBLEM STATEMENT

Each failure mode reflects single-axis reasoning about a multi-axis decision. The unified problem is that **the capex/opex inflection is a multi-parameter, workload-class-dependent, time-varying decision** that no single axis can resolve. Section 3 specifies the architecture for resolving it correctly.

3. THE WORKLOAD-CLASS TAXONOMY AND THE CROSSOVER MATH

The capacity routing in MAO-P2 partitioned workload into five classes. Each class has a distinct inflection profile.

3.1 STEADY-STATE INFERENCE

Steady-state inference is the dominant workload class in mission-ready agentic operation: sustained, predictable, high-utilization. The crossover math is straightforward. Let H be hours per month at sustained utilization, C_o the per-hour cloud opex price, C_c the amortized capex per hour (capex spread over expected lifecycle hours), and O_p the operational overhead per hour (power, cooling, ops staffing). The inflection sits where $H \times C_o = H \times (C_c + O_p)$, i.e., where $C_o = C_c + O_p$. For 2026 accelerator generation prices, the inflection for steady-state inference sits at substantially lower utilization than the cloud-default slogan suggests.¹¹¹²

3.2 BURST INFERENCE

Burst inference is the workload class for which cloud opex was originally architected and remains genuinely valuable. The elasticity discount is real: the operator pays only for the burst hours, not for the headroom. The crossover math weights utilization heavily; for genuinely bursty workloads (low average utilization, high peak), opex beats capex over any realistic operating window. The federation routes burst workload to cloud-managed substrate for this reason (MR-P9).¹³

3.3 PERSISTENT MEMORY OPERATIONS

Persistent memory operations — read/write against the authoritative tier — have a cost structure dominated by storage and egress rather than by accelerator-hour. The inflection here is sensitive to data-egress pricing and to the operator's authoritative-tier-location decision (MAO-P1, MR-P9): if the authoritative tier is on cloud-managed storage, opex dominates; if on-premises, capex dominates. The Robo Stack federation's choice to anchor the authoritative tier on Omen reflects this analysis.

11 Christina Delimitrou and Christos Kozyrakis, "Quasar: Resource-Efficient and QoS-Aware Cluster Management," *Proceedings of ASPLOS '14*, 2014, <https://doi.org/10.1145/2541940.2541941>.

12 Frank Pallas et al., "The Economics of Cloud Computing: A Survey," *Future Generation Computer Systems* 115 (2021): 78–96, <https://doi.org/10.1016/j.future.2020.08.019>.

13 Forthcoming MR-P9 *Operational Plan* (v1.0-preprint, in concurrent drafting; yks-build#99).



3.4 TRAINING AND FINE-TUNING BURSTS

Training and fine-tuning workloads are episodic and high-intensity — typically days to weeks of sustained high utilization, separated by long idle intervals. The crossover math here favors cloud opex for organizations whose training cadence is infrequent (the capex of on-prem training-class accelerators would idle most of the time) and favors capex for organizations whose training cadence is frequent enough to keep on-prem training-class accelerators utilized.¹⁴

3.5 DDIL FORWARD-DEPLOYED WORKLOAD

DDIL forward-deployed workload is the structural exception. Cognitive nodes operating disconnected from the federation core (MR-P4) cannot route to cloud opex regardless of the inflection math, because the workload's connectivity profile makes cloud-managed substrate inadmissible. The decision is not economic; it is constraint-imposed. The on-prem capex is what the architecture requires, and the inflection question does not apply to this class.¹⁵

3.6 THE AGGREGATED ROUTING IMPLICATION

The five-class taxonomy implies that operators should route each class to the substrate where its inflection favors it, not aggregate across classes and choose a single substrate. The federation pattern documented in MR-P9 is the operational form of this principle: steady-state on Omen (capex side of inflection); burst on AWS (opex side); persistent memory anchored on Omen (storage-side inflection); DDIL on forward-deployed Omen-class nodes (structural exception).

4. THE INFLECTION AS CONTINUOUS GOVERNANCE CONCERN

The inflection's parameters drift. Accelerator generations turn over every 18–24 months; regional power prices respond to grid conditions on annual cycles; sovereignty regimes change indeterminately; the operator's own scale and workload mix evolve continuously. The inflection is therefore a continuous governance concern, not a once-decided posture.

The continuous-governance instrument is the **inflection review** — a recurring sprint-aligned ceremony at which the governance layer evaluates whether current routing decisions still reflect the parameter state. The review reads curated upstream context (MAO-P1), examines current operational telemetry (MR-P9), and either confirms the existing routing posture or authorizes envelope adjustments that route workload classes differently. The cadence is sprint-aligned per Skipjack ceremony structure (P5).¹⁶

The architectural commitment is that the inflection is monitored continuously rather than reasoned about only at architecture-design time. An operator whose inflection review happens annually is asking the architecture to absorb 12 months of parameter drift between reviews; an operator whose inflection review aligns with quarterly sprint reviews is operating with much

¹⁴ Deepak Narayanan et al., “Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron-LM,” *Proceedings of SC '21*, 2021, <https://doi.org/10.1145/3458817.3476209>.

¹⁵ Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

¹⁶ Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



shorter parameter-drift exposure. The trade-off is review overhead against drift exposure; the architecture admits either choice as a governance-layer policy decision.

5. THE DDIL STRUCTURAL EXCEPTION

The DDIL-tolerant cognitive node (MR-P4) is the architecture's structural exception to the inflection analysis.¹⁷ Forward-deployed workloads operate under disconnected, intermittent, and limited connectivity conditions that make cloud-managed substrate inadmissible regardless of cost. The decision is not economic; it is constraint-imposed.

Three implications follow. First, the capex of forward-deployed cognitive nodes is not optional and is not negotiable against opex on the basis of inflection math. Second, the cognitive node's five-pillar specification (P4) constrains the substrate choice: the node must satisfy the five pillars, and not every commodity bare-metal substrate is capable of doing so without architectural work. Third, the lifecycle cost of forward-deployed nodes (refresh, maintenance, replacement) is a planning concern that ordinary cloud opex does not impose; the operator must plan for the lifecycle explicitly.

The Edge Degraded Ops Undecalogy develops the forward-deployed substrate's operational characteristics at depth. The capex/opex inflection paper acknowledges the exception and routes the analysis around it.¹⁸

6. IMPLICATIONS AND CONCLUSION

The implications of the workload-class inflection analysis are operational. Organizations should: route each workload class to the substrate where its inflection favors it (multi-policy routing per MAO-P2); review the inflection on a sprint-aligned cadence rather than at architecture-design time only; treat cost decisions as governance-layer concerns rather than infrastructure-only; and recognize the DDIL structural exception as outside the economic analysis. The core claim of this paper: **the capex/opex inflection is workload-class-dependent and time-varying; uniform application of cost-default heuristics produces systematic over-payment on the workload mix that mission-ready agentic operation produces.**

Returned to HGC³AE² letter by letter:

H — Human-governed. Cost decisions of strategic scale are governance-layer concerns. Infrastructure-only cost decisions produce locally rational, globally incoherent outcomes.

G — Curated Context. The inflection review reads curated upstream conditions (accelerator generation, power prices, sovereignty environment) plus operational telemetry from MR-P9's federation.

C¹ — Cybersecurity. Cost decisions have security implications (on-prem provenance vs cloud trust boundary); MAO-P9 develops the surface.¹⁹

¹⁷ Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

¹⁸ Forthcoming EDO-P1 *Tactical Substrate* (v1.0-preprint promotion, in concurrent drafting; yks-build#103).

¹⁹ Forthcoming MAO-P9 *Security & Provenance* (v0.1-seed, in concurrent drafting; yks-build#102).



C² — Continuity. Continuity is preserved through cost-regime transitions; workloads survive substrate refresh, accelerator generation rollover, and inflection crossings.

C³ — Coherence. All cost decisions evaluated against the same curated context, the same governance envelope, and the same workload-class taxonomy.

A — Agentially Engineered. The agentic system reports utilization, capacity availability, and inflection-relevant telemetry; humans make the inflection decisions.

E — Executed. Cost discipline is executed continuously via the inflection review cadence, not at annual budget cycle only.

The exponent — the closure — is the inflection-review loop. Operational telemetry feeds inflection re-evaluation; updated decisions update routing posture; the loop closes.

This paper closes the MAO foundation triad. The remaining MAO papers (P6 Cost Guardrails, P7 Cross-Account Harness, P9 Security & Provenance, P10 Robo Stack in Production capstone) develop operational depth on the substrate the foundation triad establishes.²⁰²¹

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20 Forthcoming MAO-P6 *Cost Guardrails* (v0.1-seed, in concurrent drafting; yks-build#100).

21 Forthcoming MAO-P10 *The Robo Stack in Production* CAPSTONE (v0.1-seed, in concurrent drafting; yks-build#91).